

Amrit Gill

Achievement 2: Real-World Applications of Machine Learning

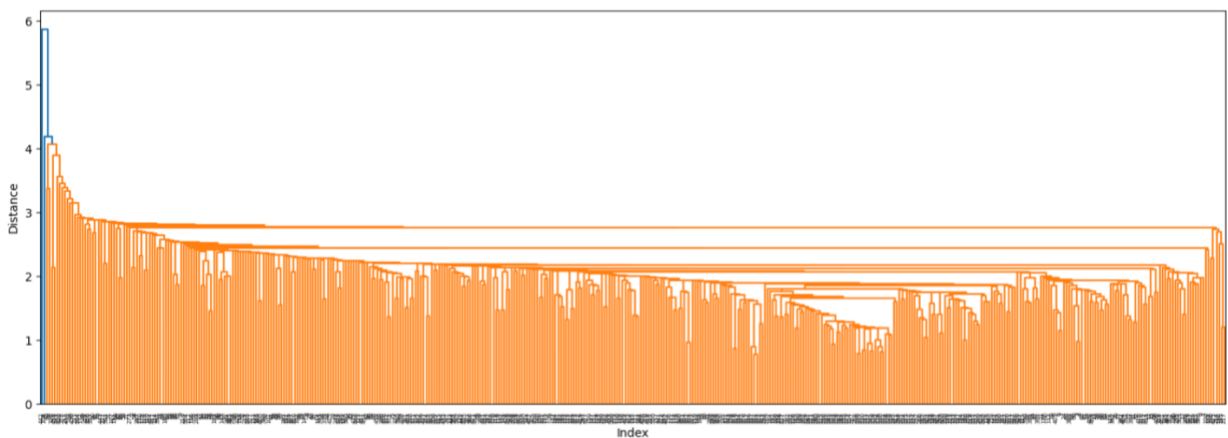
Exercise 1: Unsupervised Learning Algorithms

Dendrograms

1. Madrid vs Munich 2010

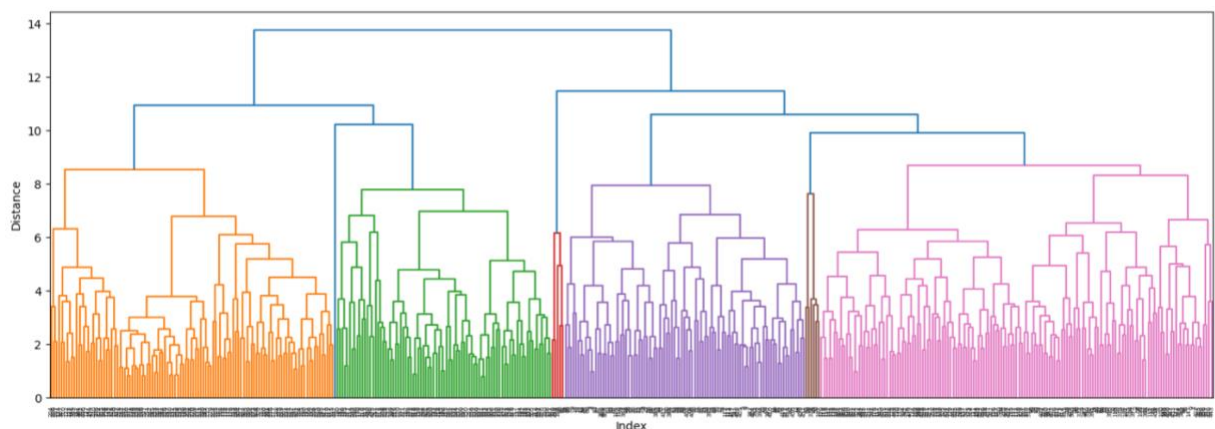
- ◆ Single Method: The distance between the two *closest* members of each cluster to determine which group they should belong to.

Dendrogram Single Method



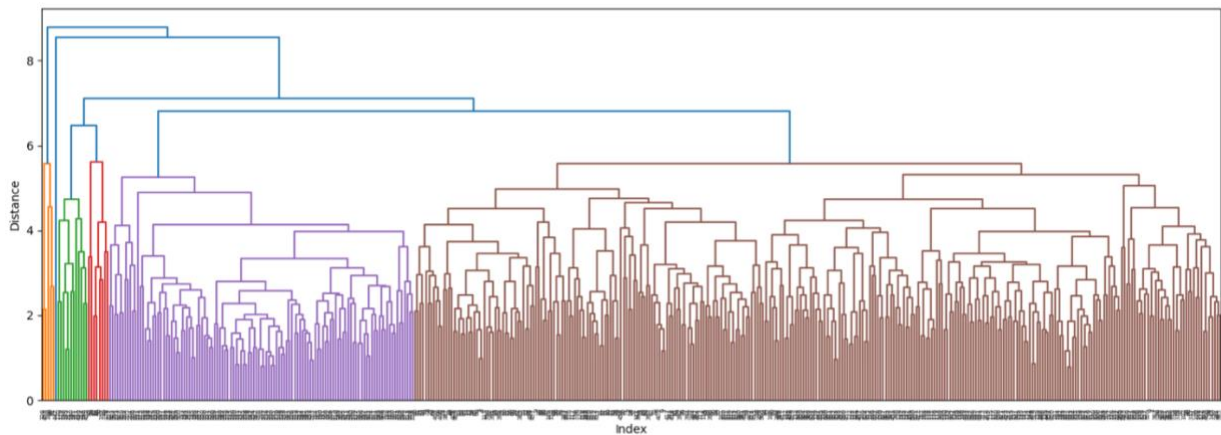
- ◆ Complete Method: The distance between the two *farthest* members of each cluster to determine which group they should belong to.

Dendrogram Complete Method



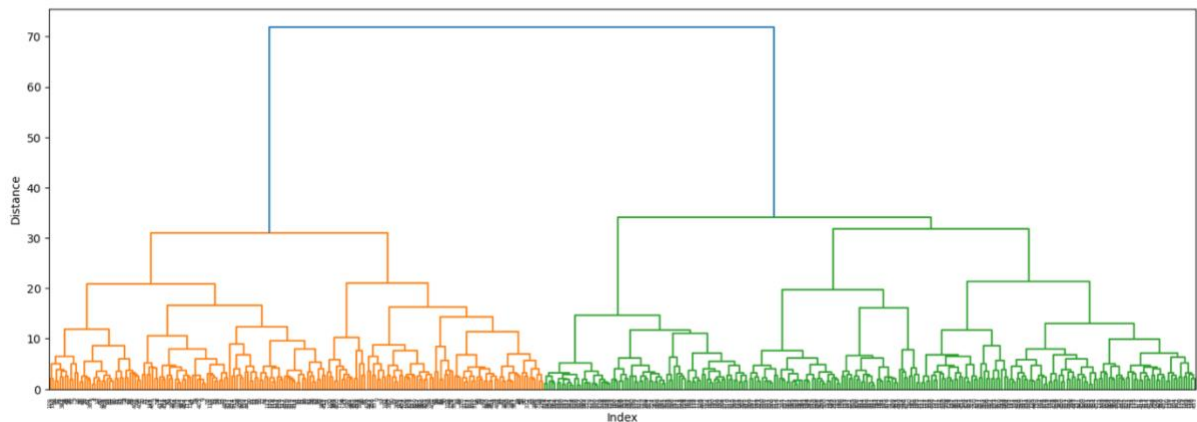
- ◆ Average Method: The distance between the *average* of the members of each cluster to determine which group they should belong to.

Dendrogram Average Method



- ◆ Ward Method: This uses a metric called the minimum increase of sum of squares (MISSQ) to find the distance between two clusters. It attempts to minimize the *variance* between the two clusters.

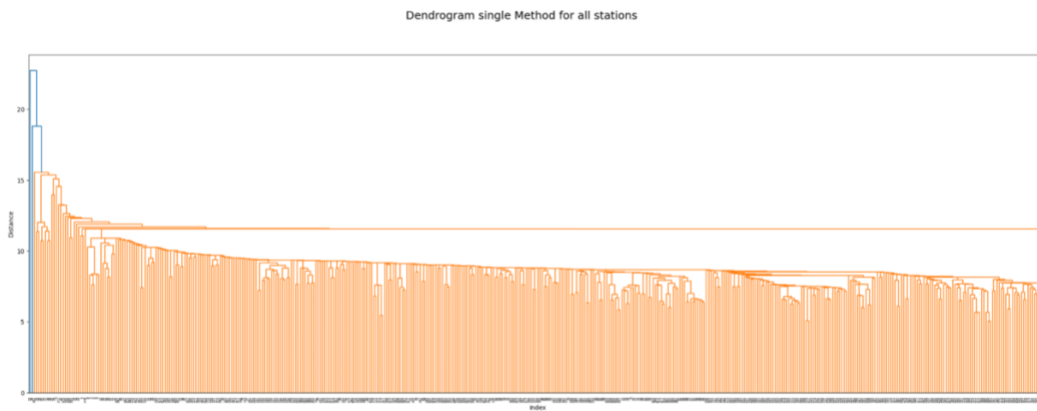
Dendrogram Ward Method



Ward method has a larger distance scale than the others. Single linkage places most days in a single category, reflecting poor cluster separation. Complete linkage forms the most well-balanced clusters, with approximately six different color-coded clusters, perhaps corresponding to four seasons or periodic trends. The Average linkage creates two big clusters and two small ones, with some distinction but less detail than Complete. Ward creates two big clusters at a higher distance, possibly reflecting geographical (e.g., Madrid vs. Munich) or seasonal (e.g., winter vs. summer) distinctions. Every method yields unique structural information: some highlight subtle fine-grained seasonal variation, while others stress broader regional contrasts or overall climatic patterns.

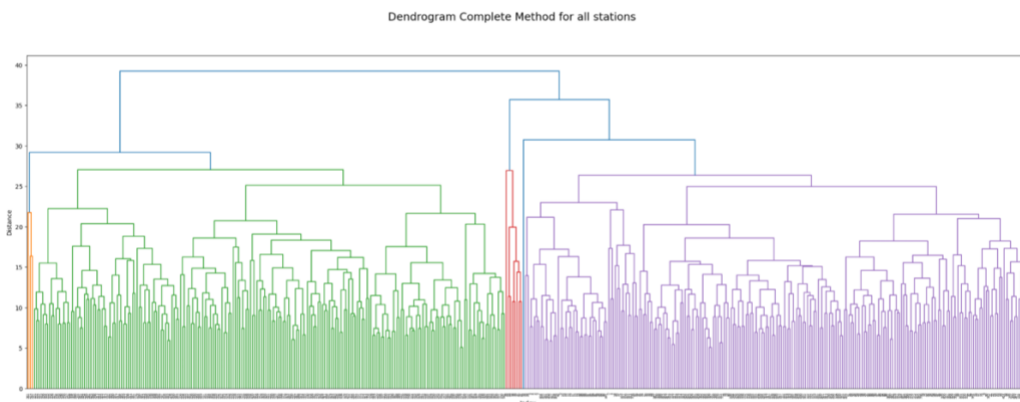
2. All Stations 2010

◆ Single Method:



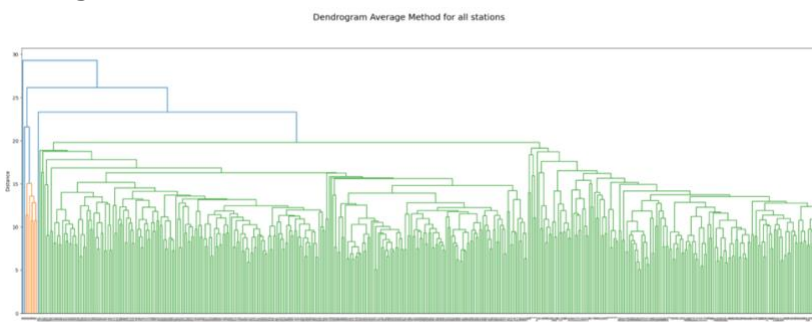
Insight: The method over-generalizes the data, grouping everything together and failing to provide any actionable or interpretable insights.

◆ Complete Method:



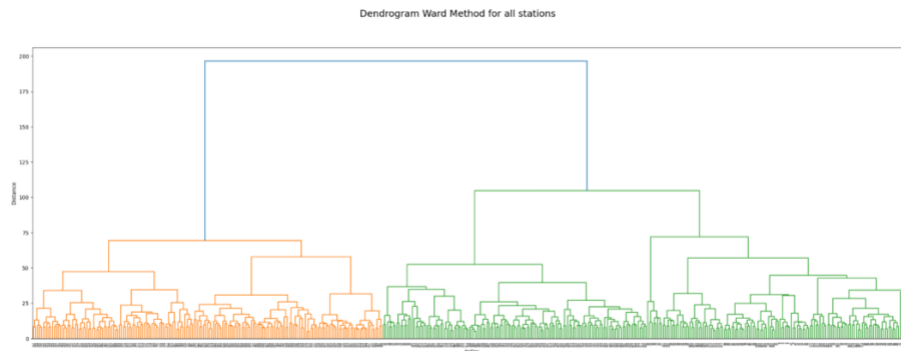
Insight: It separates into 4 categories: 2 large and 2 small, almost evenly split into two main clusters with two smaller ones.

◆ Average Method:



Insight: It's clumped into 2 groups, but the green category seems overfitted and lacks meaning, possibly due to temperature differences between hotter and cooler locations.

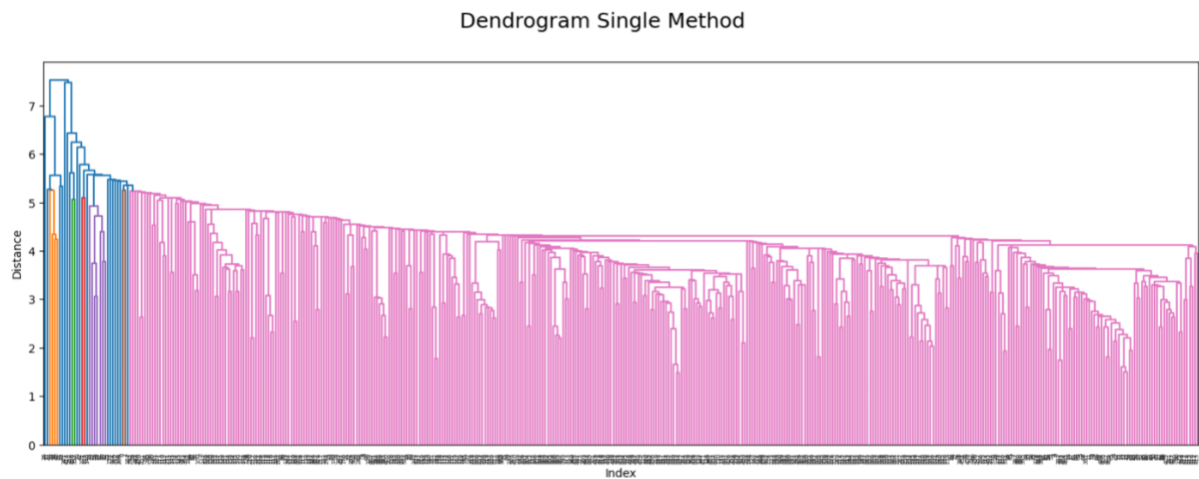
◆ Ward Method:



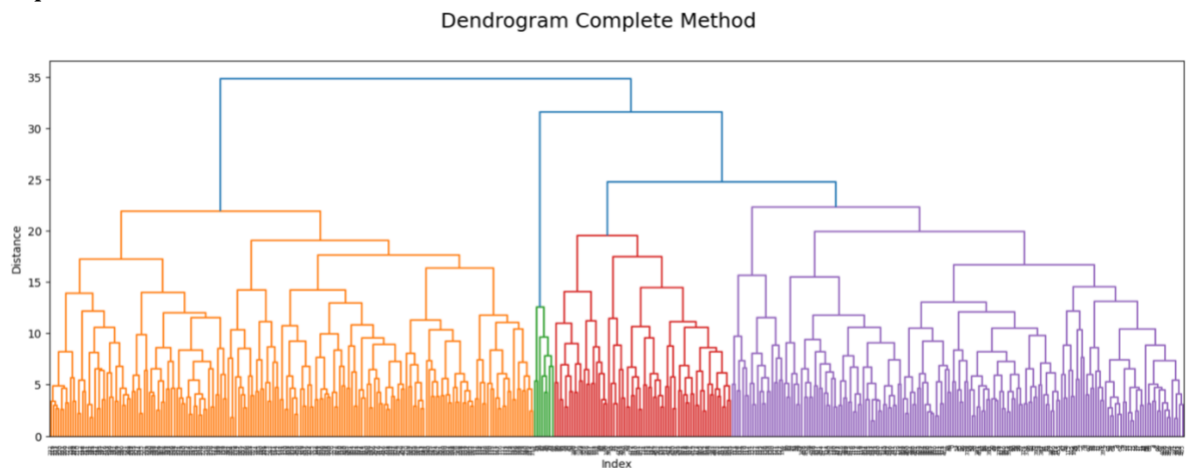
Insight: It split all locations into two clusters for 2010, likely due to location-based temperature differences.

Dendrograms with reduced data for 2010

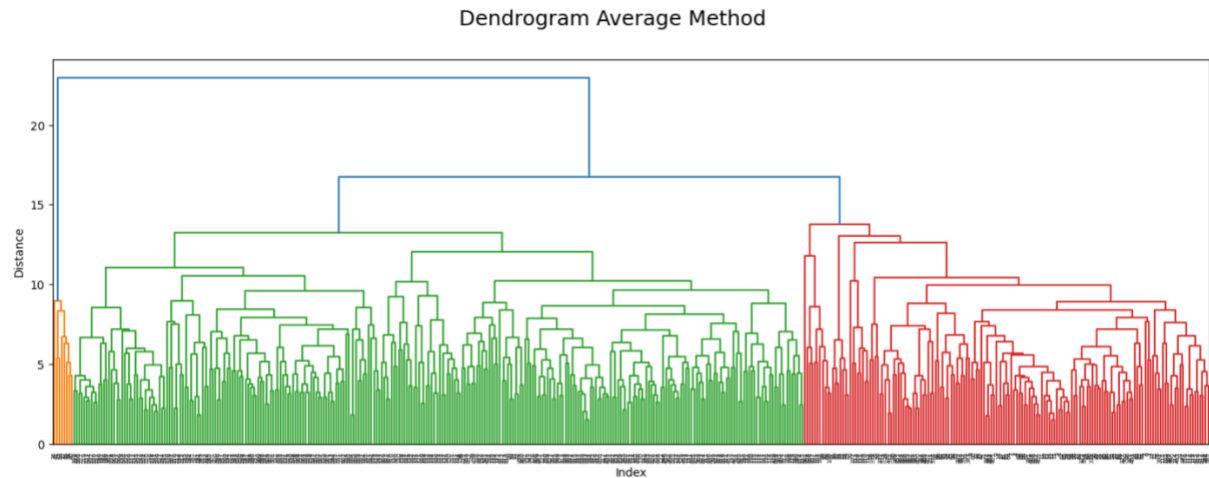
◆ Single Method



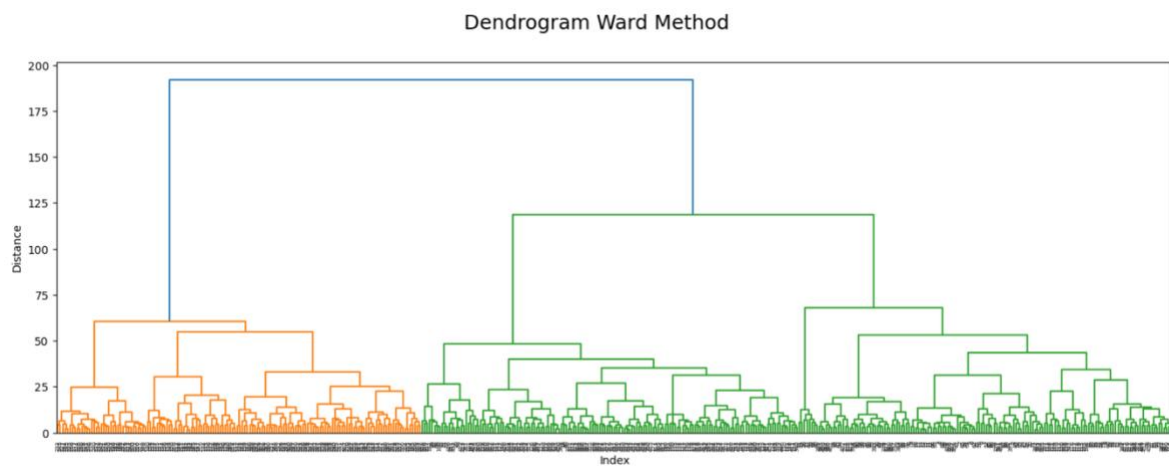
◆ Complete Method



◆ Average Method



◆ Ward Method



Summary:

1. Single Method: It shows more categories, but mostly one main cluster, so it's likely not very useful.
2. Complete Method: This shows a good mix of clusters, likely due to location or seasons.
3. Average Method: This shows three distinct clusters, which could reflect differences in location, seasonal patterns, or a combination of both, offering more meaningful separation in the data.
4. Ward Method: The Ward method results in two clear clusters, which may represent broad differences such as varying climate zones or seasonal patterns, offering a simplified but potentially meaningful division of the data.